

Assignment # 1

Economics

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CLASS: BSCS - Evening (3rd Sem)

Roll #: 93

Question No. 1

$$q_d = 28 - 4P, \quad q_s = 16 + 2P$$

Sol:-

At equilibrium point $q_d = q_s$

So,

$$28 - 4P = 16 + 2P \rightarrow 28 - 16 = 2P + 4P$$

$$12 = 6P \rightarrow P = 12/6 \rightarrow P = 2 \text{ (Equilibrium Price)}$$

Now putting $P = 2$ in q_s, q_d

$$q_s = 16 + 2(2) \rightarrow q_s = 16 + 4 \rightarrow q_s = 20 \text{ (Equilibrium Quantity)}$$

$$q_d = 28 - 4(2) \rightarrow q_d = 28 - 8 \rightarrow q_d = 20$$

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Question No. 3

$$Q_d = 20 - 5P, \quad Q_s = 4 + 3P$$

Sol.-

At equilibrium point $Q_d = Q_s$

So,

$$20 - 5P = 4 + 3P \rightarrow 20 - 4 = 3P + 5P$$

$$16 = 8P \rightarrow P = 16/8 \rightarrow P = 2 \text{ (Equilibrium Price)}$$

★ Putting $P = 2$ in Q_d & Q_s

$$Q_d = 20 - 5(2) \rightarrow Q_d = 20 - 10 \rightarrow Q_d = 10$$

$$Q_s = 4 + 3(2) \rightarrow Q_s = 4 + 6 \rightarrow Q_s = 10$$

So, 10 is Equilibrium Quantity.

★ Putt $P = 1$ in Q_d & Q_s

$$Q_d = 20 - 5(1) \rightarrow Q_d = 20 - 5 \rightarrow Q_d = 15$$

$$Q_s = 4 + 3(1) \rightarrow Q_s = 4 + 3 \rightarrow Q_s = 7$$



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Question No. 4

$$Q - 128 + 9P = 0, \quad Q + 32 - 7P = 0$$

Sol:-

$$Q_d = 128 - 9P, \quad Q_s = -32 + 7P$$

At Equilibrium point $Q_s = Q_d$

So,

$$128 - 9P = -32 + 7P \rightarrow 128 + 32 = 7P + 9P$$

$$160 = 16P \rightarrow P = 160/16 \rightarrow P = 10 \text{ (Equilibrium Price)}$$

* Putting $P = 10$ in Q_d & Q_s

$$Q_d = 128 - 9(10) \rightarrow Q_d = 128 - 90 \rightarrow Q_d = 38$$

$$Q_s = -32 + 7(10) \rightarrow Q_s = -32 + 70 \rightarrow Q_s = 38$$

So, 38 is Equilibrium Quantity

* Putting $P = 8$ in Q_d & Q_s

$$Q_d = 128 - 9(8) \rightarrow Q_d = 128 - 72 \rightarrow Q_d = 56$$

$$Q_s = -32 + 7(8) \rightarrow Q_s = -32 + 56 \rightarrow Q_s = 24$$

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Question No. 5

$$Q_d + P = 5, \quad 2P - Q_s = 5.5$$

Sol.

$$Q_d = 5 - P, \quad Q_s = -5.5 + 2P$$

At Equilibrium point $Q_d = Q_s$

So,

$$5 - P = -5.5 + 2P \rightarrow 5 + 5.5 = 2P + P$$

$$10.5 = 3P \rightarrow P = 10.5/3 \rightarrow P = 3.5 \text{ (Equilibrium Price)}$$

* Putting $P = 3.5$ in Q_s & Q_d

$$Q_d = 5 - 3.5 \rightarrow Q_d = 1.5$$

$$Q_s = -5.5 + 2(3.5) \rightarrow Q_s = -5.5 + 7 \rightarrow Q_s = 1.5$$

So, 1.5 is Equilibrium Quantity

* Putting $P = 3$ in Q_s & Q_d

$$Q_d = 5 - 3 \rightarrow Q_d = 2$$

$$Q_s = -5.5 + 2(3) \rightarrow Q_s = -5.5 + 6 \Rightarrow Q_s = 0.5$$

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<http://win10.io>

* Putting $P = 4$ in q_s & q_d

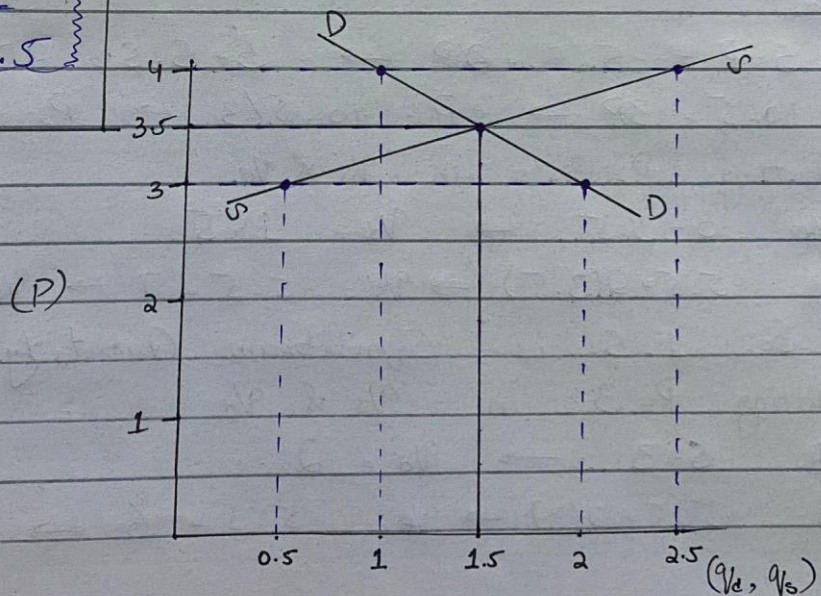
$$q_d = 5 - 4 \rightarrow q_d = 1$$

$$q_s = -5.5 + 2(4) \rightarrow q_s = -5.5 + 8 \rightarrow q_s = 2.5$$

* Schedule:-

P	3	3.5	4
q_d	2	1.5	1
q_s	0.5	1.5	2.5

* Diagram:-



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4. Solve the following demand and supply equations by algebraic method, find equilibrium price and equilibrium quantity, also draw the graph.

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a. $q_d = 28 - 4p$

$q_s = 16 + 2p$

d. $q - 128 + 9p = 0$

$q + 32 - 7p = 0$

b. $q_d = 25 - 3p$

$q_s = -10 + 4p$

e. $q + p = 5$

$2p - q = 5.5$

c. $q_d = 20 - 5p$

$q_s = 4 + 3p$

Answers:

Serial No	Equilibrium Price (Pe)	Equilibrium quantity (Qe)
a	2	20
b	5	10
c	2	10
d	10	38
e	3.5	1.5

